

SACAT: Student-Adaptive Computerized Adaptive Testing

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ABSTRACT

Computerized Adaptive Testing refers to an online student ability testing system. It consists of Selection Algorithm and Cognitive Diagnosis Module (CDM). The Selection Algorithm aims to carefully select a small set of questions for the newly arrived student to answer, and the CDM aims to infer the ability of the newly arrived student based on his answers to the selected questions. The CDM faces the challenge of inferring student ability based on the short question-answer sequence. The state-of-the-art approach employs a meta-learning framework to solve this challenge. Specifically, in the outer-layer of the meta-learning framework, it learns an average student ability from all students as the initial ability for the newly arrived student, while in the inner-layer of the meta-learning framework, it infers the newly arrived student's ability from the short sequence. However, this approach suffers from two major limitations: Firstly, collaborative information between the newly arrived student and all students was not utilized. Secondly, the meta-learning framework has a slow convergence. This paper introduces Student-Adaptive Computerized Adaptive Testing (SACAT), which addresses these issues by: Firstly, use the short sequence of the newly arrived student to retrieve similar students from all students. Secondly, design a fast adaptive function that can adaptively generate an ability representation suitable for the newly arrived student from abilities of similar students. Through extensive experiments on three real-world student response datasets, we validate that SACAT can outperform existing CAT methods. Furthermore,

through the fast adaptation, we show that SACAT achieves fast model convergence.

CCS CONCEPTS

• **Human-centered computing** → *Empirical studies in collaborative and social computing*; • **Applied computing** → **Collaborative learning**.

KEYWORDS

few-shot learning, fast adaption, Computerized Adaptive Testing

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1 INTRODUCTION

In online testing platforms (e.g. GRE, GMAT, SAT), Computerized Adaptive Testing (CAT) systems have been widely used to assess students' abilities. CAT systems show their practicability because of the capability to accurately estimate students' abilities with a reasonable amount of carefully selected questions [23]. As shown in Figure 1(a), a CAT system consists of two parts: (1) Selection Algorithm, which aims to select a small set of most informative questions for the newly arrived student to estimate his or her ability [13, 22] and (2) Cognitive Diagnosis Module (CDM), which aims to accurately infer student ability from his or her short question-answer sequence [8, 10]. These two parts work iteratively with question selection and ability inference. Current research in CAT focuses on two main aspects. On one hand, some works address the issue of how Selection Algorithm selects the most effective questions [2, 4, 12]. On the other hand, some works deal with how CDM infers student ability through short sequences, and the most effective methods [11, 21, 23] involve the use of a dual-layer structure of meta-learning. Specifically, as shown in Figure 1(b), in the outer-layer, the average ability across all students is determined. In the inner-layer, the average ability is used as the initial ability for

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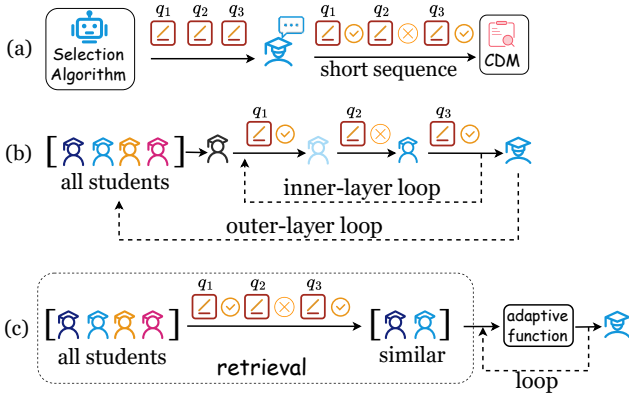


Figure 1: (a) In a CAT system, Selection Algorithm works together with Cognitive Diagnosis Module (CDM). (b) Current works learn initial student ability from historical students and infer final student ability based on the short question-answer sequence. (c) SACAT uses the short sequence to retrieve similar students and designs a fast adaptive function to guide ability inference for the newly arrived student.

the newly arrived student, and then the final ability of the newly arrived student is inferred from his short sequence. When fitting CAT in the framework of meta-learning, such as BOBCAT [11] and NCAT [23], a straightforward treatment is to copy the outer-layer student ability learned from all students as the initial ability of each newly arrived student in the inner-layer. Then the inference of the ability in the inner-layer starts from the initialization. However, such approaches have two weaknesses. Firstly, students are different and their ability varies, and these methods do not utilize collaborative information between the newly arrived student being tested and all students. Intuitively speaking, among all students, those with similar question-answer sequences to the short sequence of the newly arrived student being tested are expected to have abilities closest to the newly arrived student being tested. Using similar students in a collaborative manner instead of using all students would enhance student ability assessment. For instance, using the retrieval process shown in Figure 1(c) to retrieve the students similar to the newly arrived student being tested replaces the "all students" in Figure 1(b). This way, collaborative information is leveraged, eliminating interference from dissimilar students in the "all students" group, resulting in a more accurate initial ability estimation for the newly arrived student being tested. Secondly, as shown in Figure 1(b), the dual-layer structure alternates between optimizing the average ability of all students in the outer-layer and optimizing the individual student ability in the inner layer. During each round of optimizing the average student ability in the outer layer, some degree of interference from dissimilar students is introduced, leading to inefficient convergence of student ability inference. Based on this consideration, leveraging collaborative information and adaptive inferring ability for different students are essential.

This paper introduces Student-Adaptive Computerized Adaptive Testing (SACAT), which assists CDM in inferring student abilities

more accurately by utilizing collaborative information and designs a fast adaptation mechanism to expedite the convergence of student ability inference. Specifically, as shown in Figure 1(c), SACAT consists of two processes: the collaborative retrieval process and the fast adaptation process. It starts with the collaborative retrieval process and then moves on to the fast adaptation process. In the collaborative retrieval process, SACAT employs the short sequence of the newly arrived student to retrieve similar students from all students and calculate the abilities of these similar students. In the fast adaptation process, SACAT utilizes the abilities of similar students to infer the abilities of the target student. Through extensive experiments on three real-world student response datasets, namely Assist09, Eedi-2, and EdNet, we show proposed SACAT significantly outperforms other compared methods in the metric of prediction accuracy. To further analyze the effectiveness, we conducted a case study to show that SACAT can map different types of students into separate ability domains. Furthermore, we show that SACAT achieves fast model convergence. As a CAT system, we also show that SACAT results in acceptable question exposure and context overlap [19].

The contributions of our paper are summarized as:

- Introduction of collaborative information to enhance CDM's accuracy in inferring student ability when dealing with the short sequence. To our knowledge, SACAT is the first work that explicitly leverages collaborative information in a CAT system.
- Design of a student ability fast adaptation mechanism to expedite the convergence of CDM when utilizing collaborative information.
- Experimentally validate the state-of-the-art performance and improved convergence.

2 RELATED WORKS

2.1 Computerized Adaptive Testing

Computerized Adaptive Testing (CAT) [7] refers to an online student-ability-testing system where Selection Algorithm aims to select the most informative questions for students with different abilities and CDM aims to measure the ability of each student accurately. For a long time, classical CAT systems have used the Item Response Theory (IRT) model [9] as the CDM. Most Selection Algorithms, such as Maximum Fisher Information (MFI) [12] and Kullback-Leibler Information Index (KLI) [4], are specially designed for IRT models.

Deep learning has recently been a powerful tool for designing CAT systems. A fruitful line of Selection Algorithms prospers, such as MAAT [2], which uses active learning. At the same time, CDM evolves from IRT to MIRT [18], Neural Network (NN), and NCDM [20]. Inferring student ability from the short sequence is a few-shot learning problem, and recently some work has focused on solving similar problem [15]. The state-of-the-art methods, such as NCAT [23] and BOBCAT [11], leverage meta-learning to enhance CDM. Meta-learning adopts a dual-layer optimization framework, namely outer-layer and inner-layer optimization. In the outer-layer, the average ability across all students is determined. In the inner-layer, the average ability is used as the initial ability for the newly arrived student, and then the final ability of the newly arrived student is

inferred from his short sequence. However, merely learning average ability as the initial student ability is sub-optimal. In the view of collaborative learning, it might be helpful for CAT systems to further retrieve similar students for the newly arrived student being tested and to adaptively infer the ability of different students. The proposed SACAT leverages collaborative information. Furthermore, SACAT introduces an ability adaptive function to infer abilities for different students.

3 PRELIMINARIES

CAT Process and Problem Statements As mentioned before, a CAT system consists of two components: Selection Algorithm and CDM. In meta-learning based methods and SACAT setting, the dataset of each student is split into two parts: (1) Selection Set $\mathcal{A}$, which is used for Selection Algorithm to select questions, and (2) Query Set $\mathcal{Q}$, which is used to validate the accuracy of student ability inferred by CDM.

As illustrated in Figure 1(a), the typical process of CAT is as follows: For the newly arrived student u_i , at time step t , Selection Algorithm, which is denoted as $q_t \sim \pi(s_{t-1}^i)$, selects question $q_t \in \mathcal{A}$ for student u_i based on his short question-answer sequence s_{t-1}^i at time step $t-1$. Here, $s_{t-1}^i = \{(q_1, a_1), \dots, (q_k, a_k), \dots, (q_{t-1}, a_{t-1})\}$, and (q_k, a_k) means question q_k with its corresponding answer result a_k at time step k . After student u_i answered question q_t , we get question-answer pair (q_t, a_t) and update question-answer sequence s_{t-1}^i to s_t^i . Then the short sequence s_t^i is sent to CDM, which is denoted as $\theta_t^i \sim \mathcal{M}(s_t^i)$, to infer ability θ_t^i for student u_i at time step t .

We denote the real ability of student u_i as θ_i^* . Given the final time step T , the goal of the CAT system is to make the inferred $\hat{\theta}_i^T$ as close to the real student ability θ_i^* as possible. Our work focuses on enhancing CDM so that the $\hat{\theta}_i^T$ inferred by CDM from short sequence s_T^i at time step T gets as close to the real student ability θ_i^* as possible. To validate the discrepancy between $\hat{\theta}_i^T$ and the true student ability θ_i^* , we compute the AUC of answer result prediction $\hat{a}_l$ and the ground truth a_l for all corresponding question $q_l \in \mathcal{Q}$. This prediction procedure is formulated as $\hat{a}_l = \mathcal{G}(\hat{\theta}_i^T, q_l)$.

4 METHODS

We proposed SACAT, which consists of the collaborative retrieval process and the fast adaptation process, to help CDM leverage collaborative information to more accurately infer ability from the short question sequence and get faster convergence. Firstly, we introduce the collaborative retrieval process, which employs the short sequence of the newly arrived student to retrieve similar students from all students and calculate the abilities of these similar students. Secondly, we introduce the fast adaptation process, which utilizes the abilities of similar students to infer the abilities of the newly arrived student. Finally, we discuss the adaptive function used in the fast adaptation process.

Collaborative Retrieval Inspired by retrieval methods in other areas [14, 16, 17], we designed our Collaborative Retrieval Method. For arbitrary student u_i , his question-answer sequence at time step t is denoted as $s_t^i = \{(q_1, a_1), \dots, (q_k, a_k), \dots, (q_t, a_t)\}$. Here, $(q_k, a_k) \in s_t^i$ fulfills $q_k \in \mathcal{A}$ and $a_k \in \{-1, 1\}$. $a_k = 1$ means

student u_i answered q_k correctly, while $a_k = -1$ means incorrectly. Let $s^i \in \{-1, 0, 1\}^{|\mathcal{A}|}$ be his state. q_k -th dimension of s^i is assigned the value as $s^i[q_k] = a_k$ for all $(q_k, a_k) \in s_t^i$. For $q_l \in \mathcal{A}$ and $(q_l, a_l) \notin s_t^i$, $s^i[q_l]$ is assigned the value 0. Here, $s^i[q_l] = 0$ means question q_l is not answered by student u_i . When student u_i is the student to be tested ability, we retrieve students similar to him as follows:

Definition 4.1. Similar Student: Given student u_i with state s^i , and student u_m with state s^m , if $|\cos(s^i, s^m)| \leq \alpha$, student u_m and student u_i is similar student to each other. Here, α is a positive scalar as hyperparameter.

For each student u_m similar to student u_i , we calculate his ability $\hat{\theta}_m$ as follows:

$$\hat{\theta}_m = \mathcal{M}_{\gamma^*}(s^m) \quad (1)$$

$$s.t. \quad \gamma^* = \underset{\gamma}{\operatorname{argmin}} \sum_{q_k \in \mathcal{A}_m} \ell(a_k, \mathcal{G}(\mathcal{M}_{\gamma}(s^m), q_k)), \quad (2)$$

where γ is the parameters of CDM $\mathcal{M}(\cdot)$, $\mathcal{G}(\cdot)$ is the prediction function, ℓ is BCE loss, and $\mathcal{A}_m$ is selection set of student u_m .

Fast Adaptation From state s^i of student u_i , assuming we obtain n similar students $u_{m_1}, u_{m_2}, \dots, u_{m_n}$, and then calculate their corresponding abilities $\hat{\theta}_{m_1}, \hat{\theta}_{m_2}, \dots, \hat{\theta}_{m_n}$ using Eq. (1) and (2). We introduce an ability fast adaptive function $f_{\phi}(\cdot)$ with parameter ϕ to calculate the ability of student u_i from his similar students as follows:

$$\hat{\theta}_i = f_{\phi^*}(\hat{\theta}_{m_1}, \hat{\theta}_{m_2}, \dots, \hat{\theta}_{m_n}) \quad (3)$$

$$s.t. \quad \phi^* = \underset{\phi}{\operatorname{argmin}} \sum_{q_k \in \mathcal{Q}_i} \ell(a_k, \mathcal{G}(f_{\phi}(\hat{\theta}_{m_1}, \hat{\theta}_{m_2}, \dots, \hat{\theta}_{m_n}), q_k)), \quad (4)$$

where γ is the parameters of CDM $\mathcal{M}(\cdot)$, and $\mathcal{Q}_i$ is query set of student u_i . After fast adaptation, we get $\hat{\theta}_i$ as the initial ability of student u_i for inference. From short sequence $s_t^i = \{(q_1, a_1), \dots, (q_k, a_k), \dots, (q_t, a_t)\}$ of student u_i , we get final student ability as

$$\theta \leftarrow \theta - \eta \nabla_{\theta} \sum_{(q_k, a_k) \in s_t^i} \ell(a_k, \mathcal{G}(\theta, q_k)). \quad (5)$$

Adaptive Function The ability fast adaptive function introduced in Section 4 can adopt wide function families. This section describes how we design it.

An ability fast adaptive function (short for adaptive function) $f_{\phi}(\cdot)$ takes abilities of similar students as inputs and generates initial student ability suitable for each student as output.

When the adaptive function adopts a linear neural network, namely one-layer MLP without activation, we name the method f-linear. When the adaptive function adopts a multi-layer MLP network with activation, we name it as f-nn. The architecture we use is a two-layer MLP network with Sigmoid activation. We have tried to make f-nn module deeper and wider, but the performance does not change much.

Table 1: The AUC and ACC performance on three public datasets. The best performance is in bold, while the second-best value is underlined. “*” indicated statistically significant improvement (measured by t-test) with p-value below 0.05.

Dataset	ASSIST09						EdNet						Eedi-2					
CDM	MIRT			NN			MIRT			NN			MIRT			NN		
Metric	AUC																	
Step	1	5	10	1	5	10	1	5	10	1	5	10	1	5	10	1	5	10
Random	68.68	68.76	70.80	69.48	70.52	71.12	73.59	74.29	74.80	69.37	70.61	71.56	69.29	73.00	74.08	69.80	73.67	75.24
MAAT	68.91	73.12	<u>75.54</u>	69.67	70.85	71.98	73.15	73.31	75.50	73.50	74.63	75.35	69.81	73.38	75.08	70.52	74.33	75.88
BOBCAT	<u>70.22</u>	<u>73.36</u>	73.76	69.61	71.98	72.72	<u>74.33</u>	<u>75.98</u>	<u>76.73</u>	<u>74.31</u>	<u>75.79</u>	<u>76.60</u>	<u>71.34</u>	<u>75.35</u>	<u>76.69</u>	<u>71.41</u>	<u>75.12</u>	<u>76.69</u>
NCAT	69.51	72.65	73.99	70.57	72.53	73.08	73.90	75.06	75.52	74.16	75.06	75.52	71.32	74.71	75.51	71.11	74.49	75.98
SACAT	73.39*	76.68*	77.26*	71.14*	75.24*	76.62*	74.44*	76.15*	76.76	74.47*	76.22*	76.84*	71.62*	75.51*	76.81*	71.46	75.48*	76.97*
Metric	ACC																	
Step	1	5	10	1	5	10	1	5	10	1	5	10	1	5	10	1	5	10
Random	67.22	67.33	68.22	<u>68.67</u>	69.02	69.33	70.75	71.13	71.39	67.43	67.78	68.47	64.38	67.07	68.35	64.63	67.50	68.64
MAAT	67.33	<u>70.64</u>	<u>73.26</u>	67.85	69.15	69.68	70.58	70.66	71.86	70.71	71.74	71.78	64.62	67.31	68.84	65.09	67.99	69.27
BOBCAT	<u>68.06</u>	70.17	70.53	67.33	68.80	68.81	<u>71.06</u>	<u>72.10</u>	<u>72.52</u>	<u>71.14</u>	<u>71.98</u>	<u>72.42</u>	65.69	<u>68.60</u>	<u>70.04</u>	65.67	<u>68.71</u>	<u>70.06</u>
NCAT	67.26	70.15	70.89	68.42	<u>70.07</u>	<u>70.22</u>	70.96	71.58	71.83	71.04	71.58	71.81	<u>65.92</u>	68.40	69.13	<u>65.71</u>	68.25	69.44
SACAT	70.94*	73.55*	74.02*	68.82	71.77*	71.82*	71.22*	72.20*	72.56	71.26*	72.04*	72.60*	66.11*	69.01*	70.28	65.87*	69.22*	70.36*

Table 2: Datasets statistics.

Dataset	Students	Questions	Interactions
ASSIST09	2.3K	26.7K	325K
Eedi-2	5K	1K	1.4M
EdNet	312K	13K	76M

5 EXPERIMENTS

This section presents the experimental settings and corresponding results in detail. The implementation will be publicly available¹. We use the following research questions (RQs) to lead experiments and discussions.

- **RQ1:** Does SACAT outperform the baseline methods on student performance prediction?
- **RQ2:** Does SACAT capture student ability more accurately than other methods?
- **RQ3:** How does the adaptive function make it effective in mapping different types of students into separate domains of student ability distribution?
- **RQ4:** Does SACAT converge better than meta-learning based method?
- **RQ5:** Does fast adaptation perform better than inner-layer optimization of the meta-learning base method?
- **RQ6:** How does SACAT behave on the aspects of question exposure and test overlap?

5.1 Experimental Settings

We now detail both quantitative and qualitative experiments. We tested three real-world student response datasets to validate SACAT’s effectiveness.

5.1.1 Datasets. We use three publicly available benchmark datasets: ASSIST09², Eedi-2³, and EdNet⁴. As shown in Table 2,

¹The code is available in supplementary materials.

²<https://sites.google.com/site/assistentmentsdata/home/assistent-2009-2010-data>

³<https://eedi.com/projects/neurips-education-challenge>

⁴<https://github.com/rriid/ednet>

we list the number of students, the number of questions, and the number of interactions. We provide preprocessing details and additional background on each dataset in the supplementary material.

We perform 5-fold cross-validation for all datasets; for each fold, we allocated 60% students for training, 20% students for validation, and 20% for testing. For SACAT, we partition the questions responded to by each student u_i into the selection set ($\mathcal{A}_i$, 80%) and the query set ($\mathcal{Q}_i$, 20%).

5.1.2 Evaluation Methods. We use both accuracy (ACC) and the area under the receiver operating characteristics curve (AUC) on the query set $\mathcal{Q}_i$ as metrics to evaluate the performance of all methods on predicting binary-valued student responses.

5.1.3 Compared methods. As mentioned before, CAT consists of Selection Algorithm and CDM. Our experiment mainly involves two classic CDM Models: Traditional item response theory MIRT [1] and recently emerging Neural Network (NN) [3]. We compare SACAT with the following CAT systems:

- **Random:** The random selection strategy.
- **MAAT** [2]: It proposes an active learning-based method, which measures questions’ informativeness by calculating the Expected Model Change (EMC) caused by each question.
- **NCAT** [23]: It is a reinforcement learning-based method, which designs an attention-based DQN to select questions.
- **BOBCAT** [11]: It’s the first bilevel optimization framework, which introduces MAML to a CAT system and adopts an approximate gradient estimate method for CAT to learn a data-driven selection algorithm.

5.1.4 Hyper-parameters. We train MIRT models using logistic regression with L2-norm regularization. For NN, we use a two-layer, fully connected network (with 256 hidden nodes, ReLU nonlinearity, 20% dropout rate, and a final sigmoid output layer) as the response model, with a student-specific, 256-dimensional ability vector as input. We implement all methods in PyTorch.

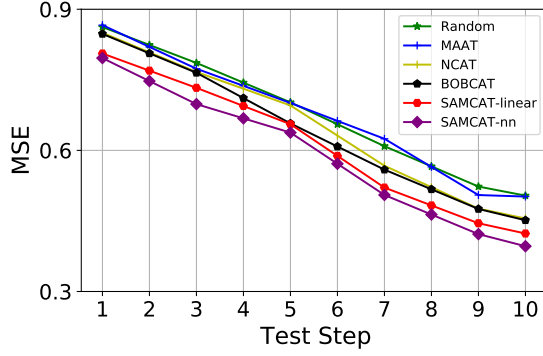


Figure 2: MSE w.r.t as test step increases on Eedi-2 dataset.

5.2 Performance Comparison (RQ1)

The student performance prediction task is generally used to evaluate the effectiveness of a CAT system. Table 1 reports the mean AUC and ACC metric on query sets of different methods after t -th step of selection, and we show the results in 1, 5, and 10 of the number of selected questions (or test length) respectively.

We have the following observations: SACAT achieves the best performance on the AUC and ACC over three benchmark datasets and in all test lengths, significantly outperforming the SOTA methods. It hints that our proposed framework SACAT has captured more accurate student ability status θ_i , so that student performance prediction in query set is more accurate.

5.3 Ability Estimation (RQ2)

A CAT system aims to use as few questions to get as accurate information about students' abilities as possible. So the estimated student ability $\hat{\theta}_i$ can be used to score the CAT system. We conduct an experiment on the Eedi-2 dataset using the squared error between the current ability parameter estimate $\hat{\theta}_i^{(n)}$ and the true ability θ_i^* as the evaluation metric. Here n means ability $\hat{\theta}_i^{(n)}$ is estimated after n questions selected by Selection Algorithm. Since the true student ability is unknown in real student response datasets, we use the ability value estimated from all questions each student responded to as a substitute. Using NN as CDM, we compare the following CAT methods: Random, MAAT, BOBCAT, NCAT, SACAT with linear adaptive function (SACAT-linear), and SACAT with nn adaptive function (SACAT-nn). Figure 2 shows the ability estimation error (averaged over five folds) for different numbers of questions selected. We see that the MSE between the model estimated ability and the true ability decreases with test steps increases, and both SACAT-linear and SACAT-nn have lower MSE compared with other methods. On one hand, SACAT achieves more accurate estimation with the same number of questions. On the other hand, SACAT needs fewer questions to get the same accuracy, which means SACAT can significantly reduce the test length.

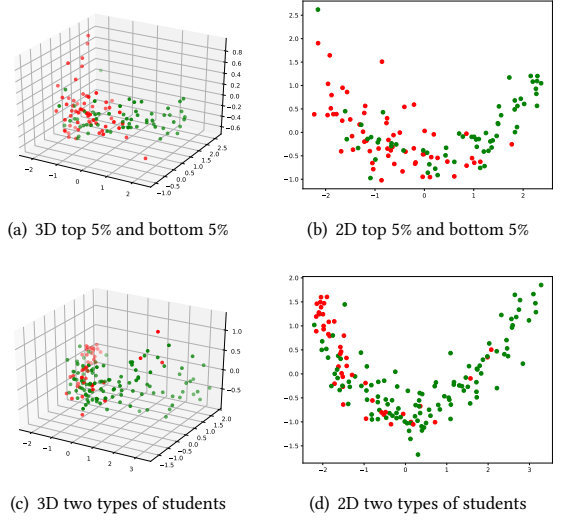


Figure 3: The students' abilities distribution after the ability function mapping on the Eedi-2 dataset. (a) 3D picture of top 5% and bottom 5% students. (b) 2D picture of top 5% and bottom 5% students. (c) 3D picture of two types of students. (d) 2D picture of two types of students.

5.4 Mapping Effectiveness of adaptive function (RQ3)

To gain deeper insights into how the ability adaptive function leads to more accurate question prediction, we take a close look at the distribution of student abilities after mapping. We experiment on dataset Eedi-2 and use nn as ability adaptive function under the SACAT framework.

Firstly, we color students whose right answers number gets top 5% as green and color students whose right answers number gets bottom 5% as red. 'Green' and 'red' sum up to about 150 students. Different types of students are mapped to separated areas of ability high dimensional space. Since student ability is a high dimensional vector, we use tSNE to figure student ability in 2D and 3D dimensions. After the ability adaptive function mapping, student abilities of Type green and Type red gather separately (Figure 3(a) and 3(b)).

Secondly, to further see whether the ability adaptive function maps students with different abilities to different areas, we choose two kinds of different students as follows. Specifically, we mark students who answer correctly from question 1 to question 4 as Type green and mark students who wrongly answer all questions as Type red. After the ability adaptive function mapping, as shown in Figure 3(c)(d), we can see Type green and Type red gather more separately than Figure 3(a)(b).

The result is that different kinds of students with different student abilities can be mapped to different areas (Figure 3(a)(b)).

5.5 Fast Adaptation (RQ4)

We use the number of training epochs to evaluate model convergence. In Table 3, we report the epochs of BOBCAT, SACAT-linear, and SACAT-nn when the model converges on the ASSIST09 dataset.

Table 3: Epochs for the different methods with different questions selected number to convergence on the ASSIST09 dataset.

n_query	1	3	5	10
BOBCAT	276 ± 54	527 ± 51	542 ± 44	448 ± 45
SACAT-linear	257±43	276 ± 23	534 ± 33	352 ± 17
SACAT-nn	254 ± 41	273 ± 41	439 ± 51	224 ± 36

Table 4: AUC of different methods at different inner level steps on the ASSIST09 dataset.

BOBCAT	SACAT-linear			
step 4	step 1	step 2	step 3	step 4
71.78	73.57	74.88	74.93	75.08

Table 5: Question exposure and test overlap rates for the different methods using NN as CDM on the Eedi-2 dataset.

Method	Exposure (median)	Exposure (> 20%)	Overlap (mean)
Random	1.02%	0.00%	1.61%
MAAT	0.61%	0.11%	4.90%
BOBCAT	0.10%	0.42%	8.54%
NCAT	0.00%	1.79%	30.00%
SACAT-nn	0.00%	0.62%	12.67%

We randomly generate seed numbers 20 times on each pair of datasets and queries. The epochs of SACAT-linear and SACAT-nn are smaller than the epoch of BOBCAT. We see that both SACAT-linear and SACAT-nn have faster convergence than BOBCAT.

5.6 Inner-level Efficiency (RQ5)

We make further analysis of the effectiveness of fast adaptation by varying the number of updating steps. We compare the fast adaptation optimization of SACAT-linear and inner-layer optimization of BOBCAT by training for a specified number of steps. Here we use NN as the CDM. Results of AUC are reported in Table 4. Regardless of the number of steps, SACAT-linear consistently outperforms BOBCAT. We can see that even SACAT-linear using 1 step achieves higher AUC compared with BOBCAT using 4 steps, which indicates SACAT has the capacity for fast adaptation.

5.7 Question Exposure and Context Overlap (RQ6)

We follow the method to measure the exposure rate distribution-chang1999stratified, and the method introduced to measure context overlapChen et al. [6]. The smaller the exposure and overlap, the better method. The details of these methods are as follows: Chang and Ying [5] proposed a χ^2 statistic to measure the skewness of the exposure rate distribution: $\chi^2 = \sum_{k=1}^n \frac{(er_k - L/n)^2}{L/n}$, where er_k is the observed exposure rate for the k -th question, L is the test length, and n is the total number of questions in the bank. Chen et al. [6] introduced a way to measure context overlap: $\hat{T} = \frac{L}{n} er_k^2 + \frac{n}{L}$.

In Table 5, we provide summary statistics on the question exposure rate and the test overlap rate on the Eedi-2 dataset. We see that

SACAT-nn results in a lower question exposure rate than existing CAT methods, and leads to an acceptable portion of overexposed questions (more than the recommended limit of 20%) [19]. On the other hand, SACAT-nn results in an acceptable test overlap rate between BOBCAT and NCAT.

6 CONCLUSIONS AND FUTURE WORK

In this paper, we proposed SACAT: Student-Adaptive CAT, which retrieves similar students and proceeds the fast adaptation process. Through extensive experiments on three real-world student response datasets, we validate that collaborative information is the important ingredient that was neglected in the BOBCAT and NCAT approaches, which can outperform existing CAT methods at reducing test length. In the future, we are committed to designing more effective ability adaptive functions depending on CDMs and datasets. As for how to choose the ability adaptive function, we leave it as future work.

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